EDS 212: Day 1, Lecture 2

Functions, graphs, and linear functions

August 3rd, 2026

Functions

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For example, $y = 3x - 5$ is a function that tells us the **value of y** at **any value of x**. In this scenario, we would probably say **y is a function of x**.

Could you also rewrite it and say **x is a function of y**?

Here, with no knowledge of what's an input and what's an output, sure - but usually in environmental data science we specify the input variable(s), and the output variable(s) carefully.

What follows is the expression in the format of:

"**[output variable(s)] is/are a function of [input variable(s)]**".

Also: output variable = **dependent variable** and input variable = **independent variable**.

Thinking about inputs and outputs

For the following combinations of related variables, which do you expect would be the **input** and the **output** in a function describing how they are related? E.g. “Evapotranspiration is a function of air temperature.”

✏️ Write your answer individually as **a sentence**.

1. fuel (biomass) / slope / wildfire severity / windspeed / air temperature
2. wind speed / power generated by wind turbine
3. soil C:N ratio / bacterial biomass / soil water content / leaf litter decomposition rate

Function notation

✏️

Function notation

Single variable (univariate) function:

$f(x) = [expression\ containing\ x]$

Multivariate function:

$g(a, T, z) = [expression\ containing\ a, T,\ and\ z]$

Evaluating functions

We evaluate functions by plugging in values of the input variables.

✏️ Example:

Evaluate $g(x, t) = 2.4x + 0.5t^2$ at $x = 3$ and $t = 10$

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✏️ Example:

Evaluate $g(x, t) = 2.4x + 0.5t^2$ at $x = 3$ and $t = 10$

$$g(3, 10) = 2.4(3) + 0.5(10^2) = 57.2$$

Graphs

Why graphs?

Graphs are a way for us to more easily process trends or patterns that may be more challenging to understand in a table or list.

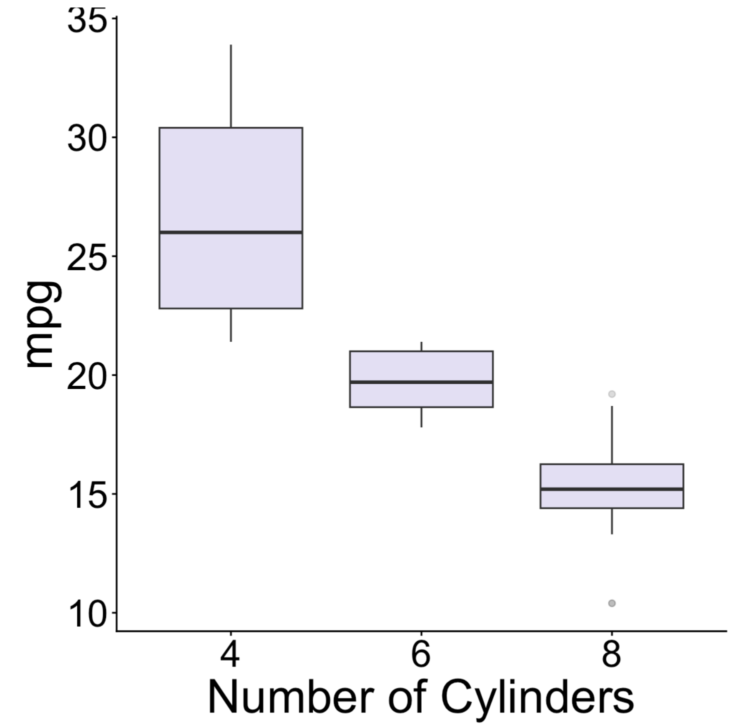
Which looks better and is easier to understand?

Show 4 entries Search:

	mpg	cyl	disp	hp
Mazda RX4	21	6	160	110
Mazda RX4 Wag	21	6	160	110
Datsun 710	22.8	4	108	93
Hornet 4 Drive	21.4	6	258	110

Showing 1 to 4 of 32 entries

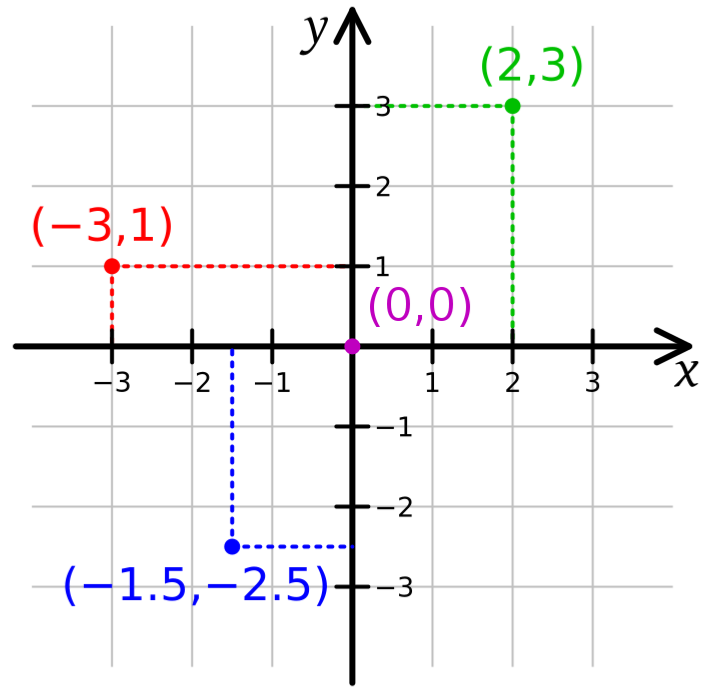
Previous 1 2 3 4 5 … 8 Next



The xy coordinate system

Graphs generally move in a 2-dimensional plane with a coordinate system using two axes:

- x-axis (horizontal axis)
- y-axis (vertical axis)



Axes units must be defined depending on the application.

We use pairs of numbers to place data:

- A point on the xy plane with **coordinates** (a, b) means the point is located at a on the x-axis and at b on the y-axis.
- Where would (2,-1) go on the graph?

We can join series of points to make graphs

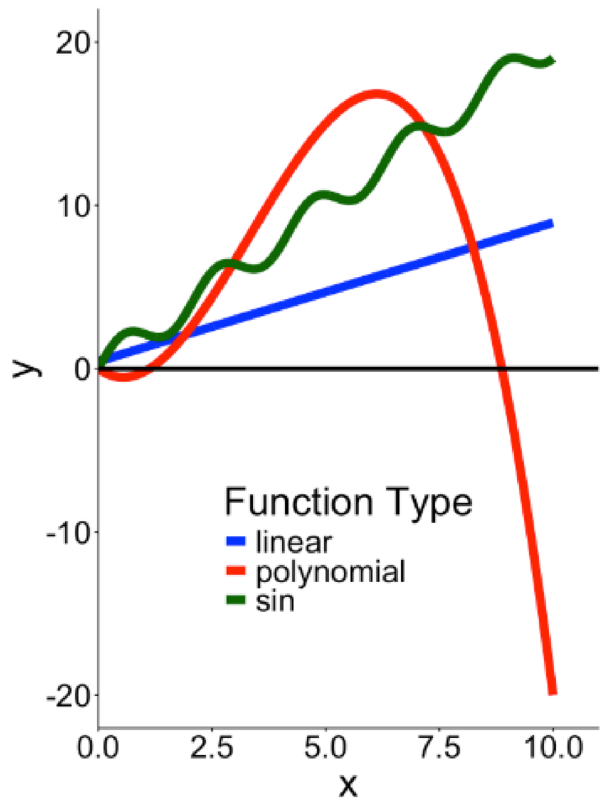
Functions on a single variable and a single output can be graphed

Two key ingredients to graphs

Intercepts

- x-intercept
  - Where does the graph intersect the x-axis?
  - In other words, for what values of x do we have $(x, 0)$ be part of the graph?

What are the intercepts of the polynomial function in red?

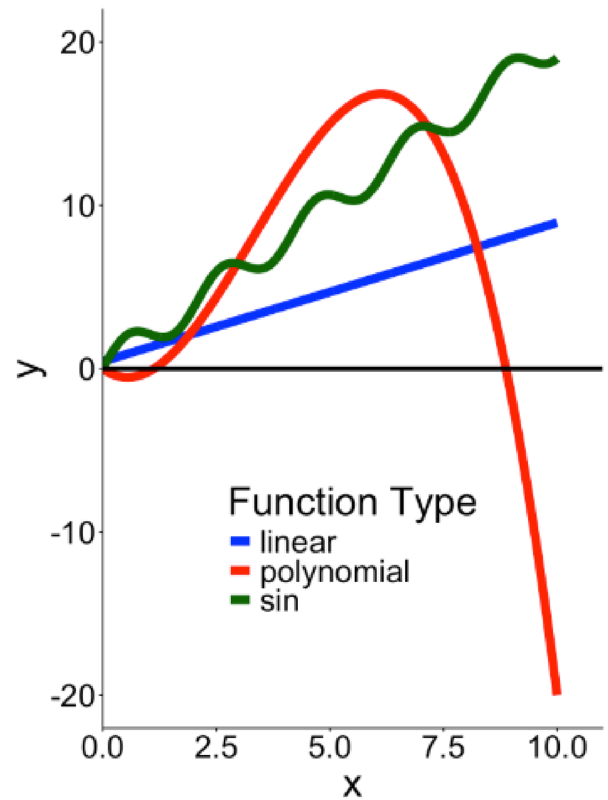


Two key ingredients to graphs

Intercepts

- x-intercept
  - Where does the graph intersect the x-axis?
  - In other words, for what values of x do we have $(x, 0)$ be part of the graph?
- y-intercept
  - Where does the graph intersect the y-axis?
  - In other words, for what value of y do we have $(0, y)$ be part of the graph?

What are the intercepts of the polynomial function in red?



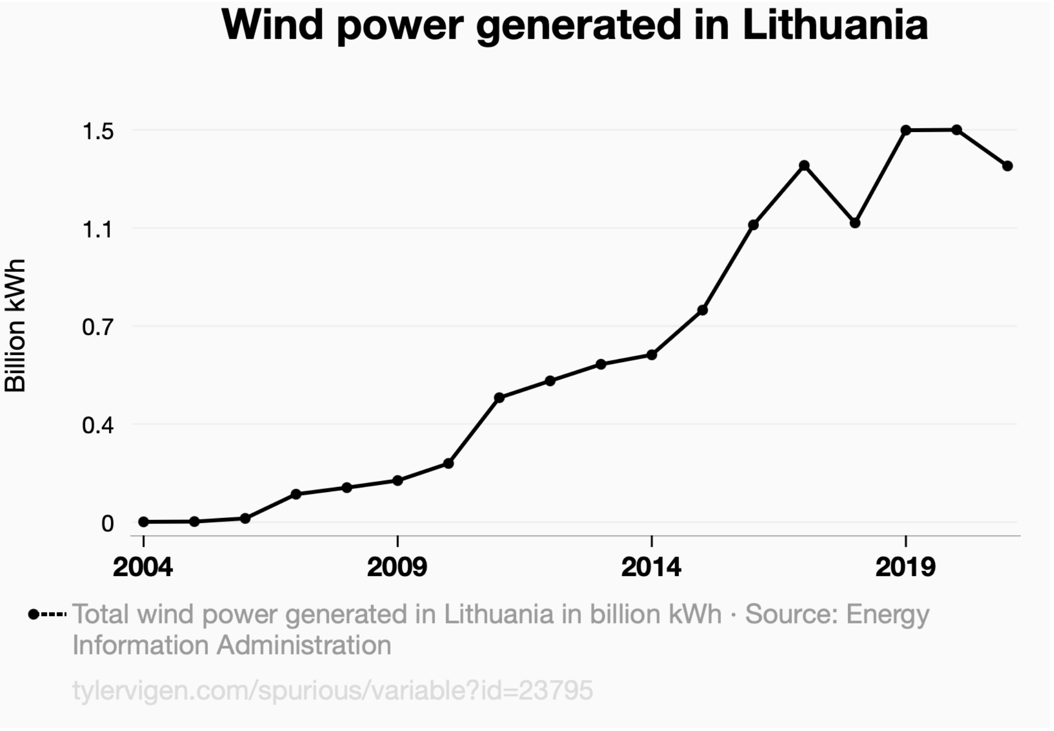
Understanding graphs

When you look at graphs, the first things you should ask:

- What variables are being plotted (e.g. x- & y-axis, including units)?
- What values are plotted (e.g. raw values, transformed, means, etc.)?
- What are the overall takeaways and am I understanding them responsibly?

Reading graphs (in general)

> "On the x-axis we have [x-variable] measured in [units] and on the y-axis the the [y-variable] measured in [units]. This figure shows the [change/pattern/relationship] between [x-variable] and [y-variable]. Overall [overall statement of pattern / trend / findings]."



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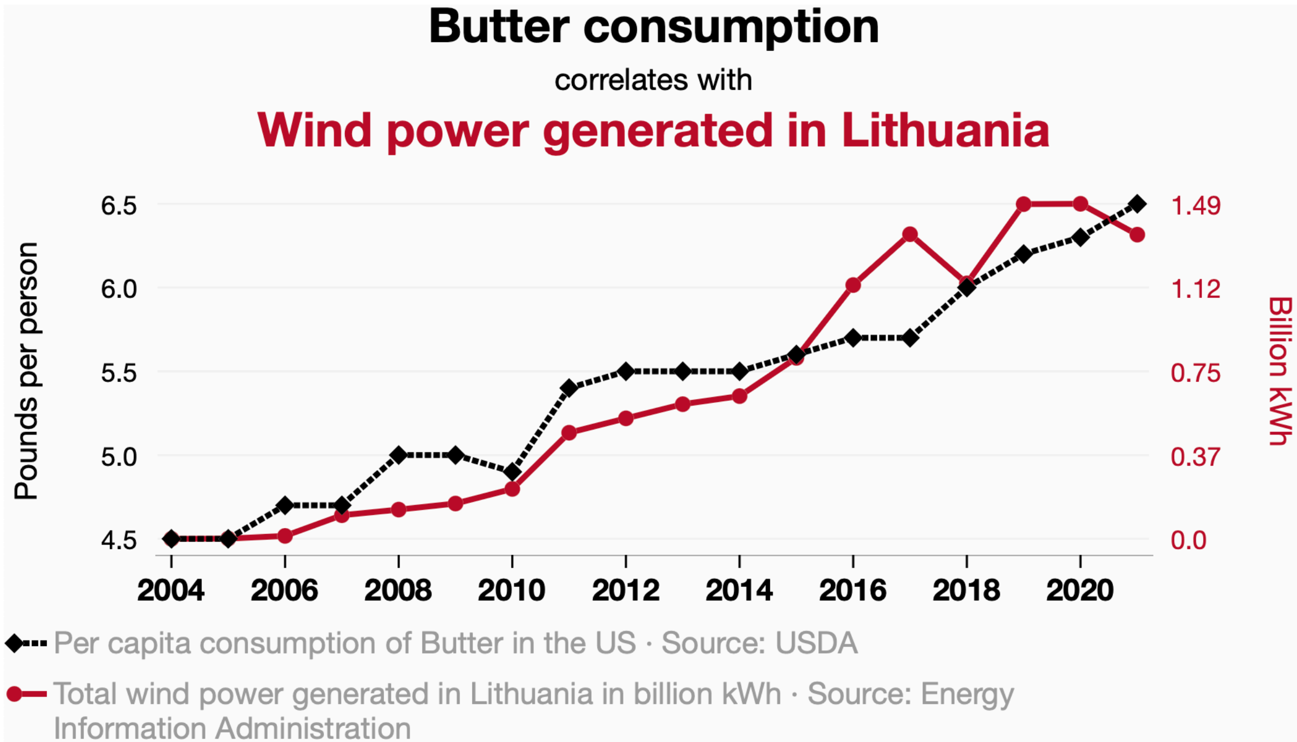
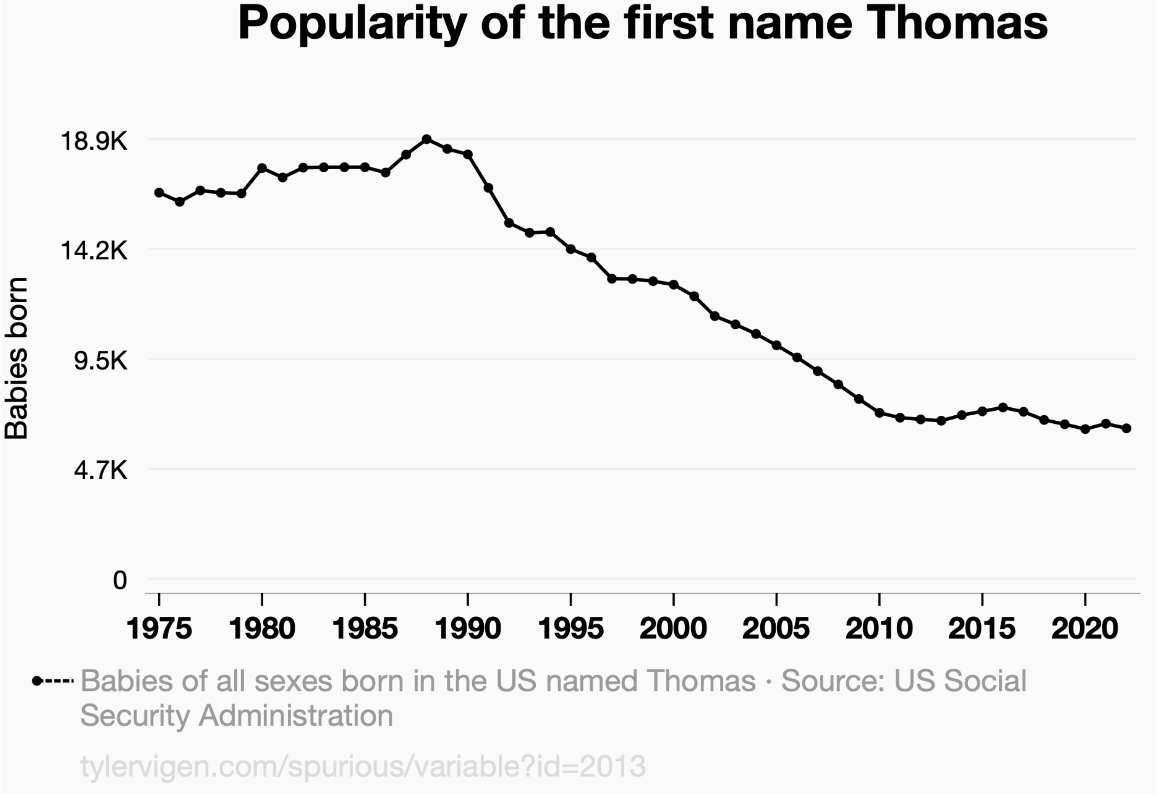


Source: TylerVigen.com

Reading graphs practice

💬 Partner 1 gets to read the graph for the first variable.

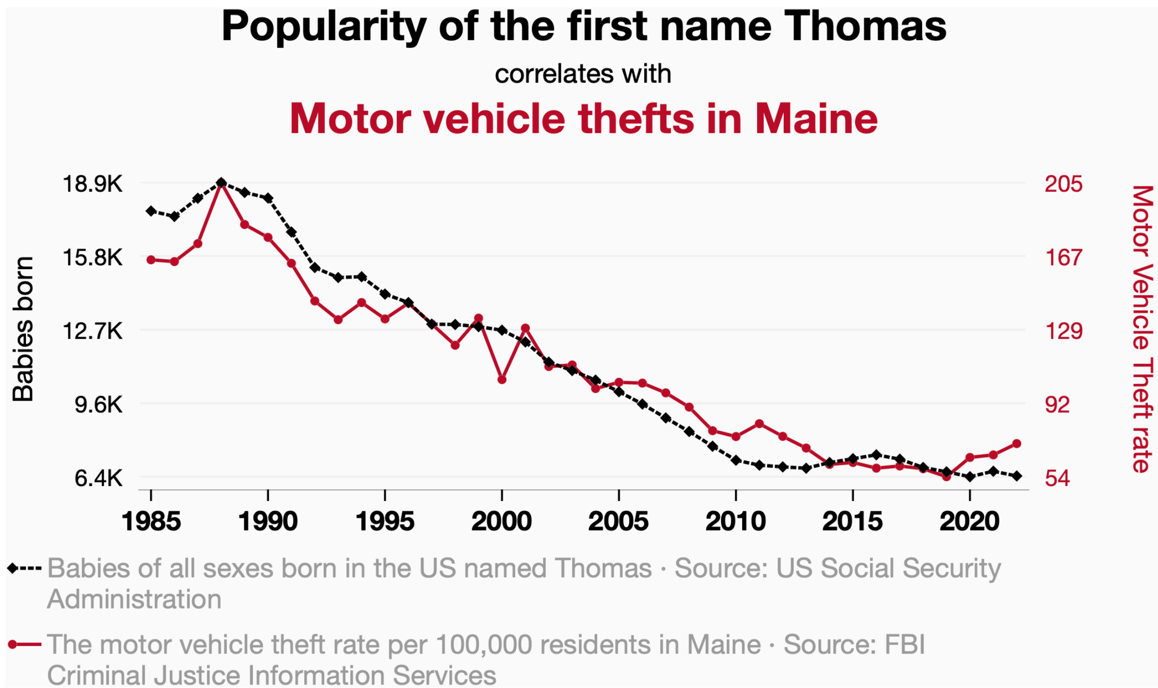
> "On the x-axis we have [x-variable] measured in [units] and on the y-axis the the [y-variable] measured in [units]. This figure shows the [change/pattern/relationship] between [x-variable] and [y-variable]. Overall [overall statement of pattern / trend / findings]."



Reading graphs practice

💬 Partner 2 gets to read the plot for the second variable, provide context for the data, and hypothesise an explanation.

> "On the x-axis we have [x-variable] measured in [units] and on the y-axis the the [y-variable] measured in [units]. This figure shows the [change/pattern/relationship] between [x-variable] and [y-variable]. Overall [overall statement of pattern / trend / findings]."



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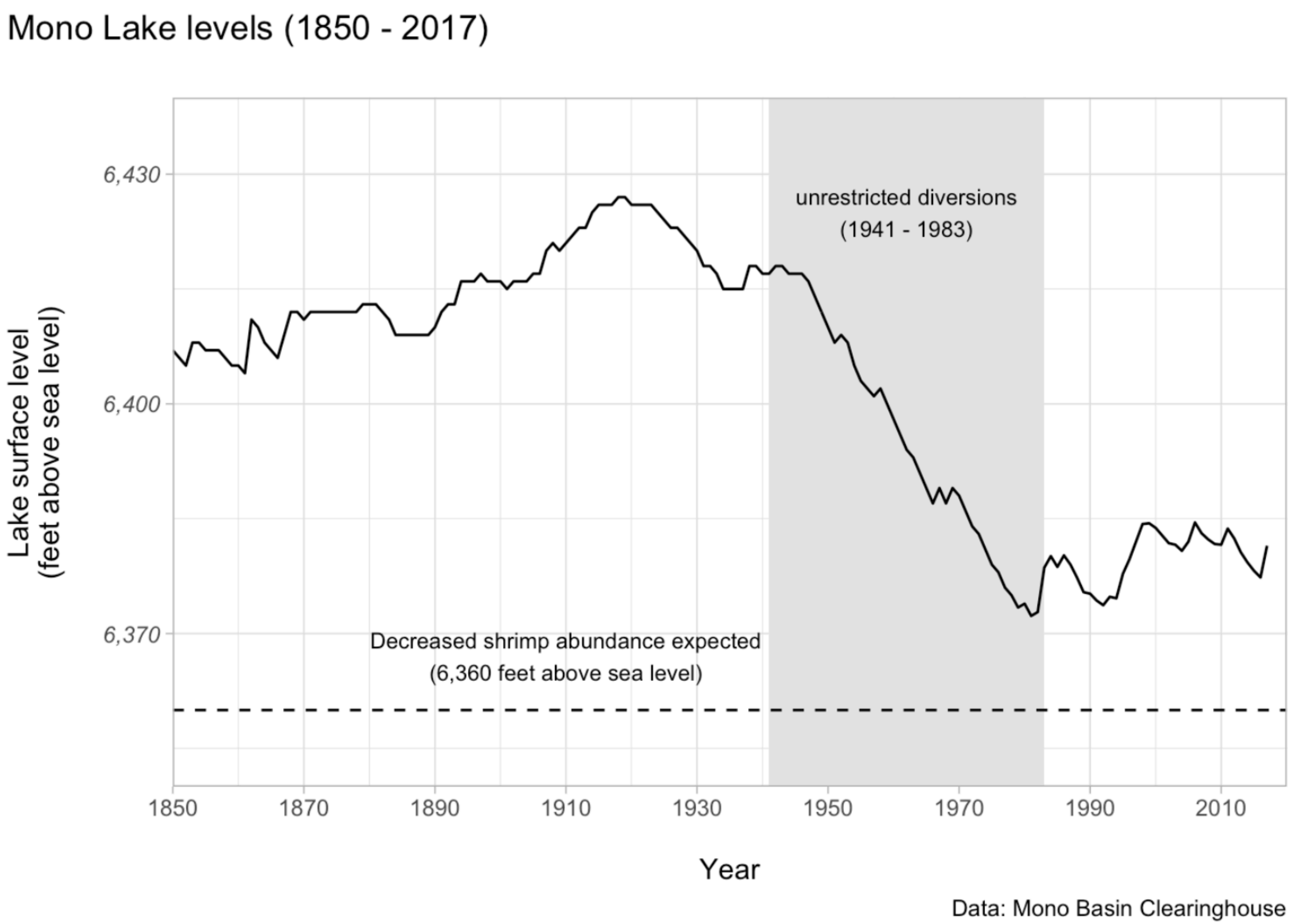
Reading graphs practice



Let's take a 5 minute break

image: Flaticon.com

Linear functions

Linear functions: slope-intercept form

✏️

Linear functions: slope-intercept form

Easiest model to describe linear relationship between two variables!

$$\underbrace{y}_{\text{output}} = \overbrace{m}^{\text{Slope}} \underbrace{x}_{\text{input}} + \overbrace{b}^{y \text{ intercept}}$$

Finding the equation of a line

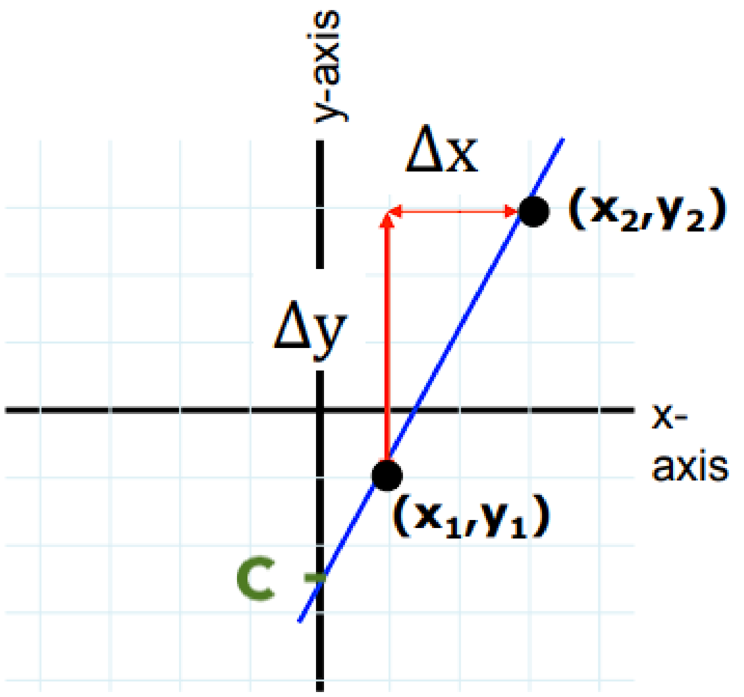
✏️

Finding the equation of a line

Given two points (x_1, y_1) and (x_2, y_2) on the line:

1. Find m, the **slope**, by using the two points:

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}.$$



2. Find c, the **y-intercept**:

Since (x_1, y_1) is on the line, it satisfies:

$$y_1 = mx_1 + c.$$

We can solve for c to obtain:

$$c = y_1 - mx_1.$$

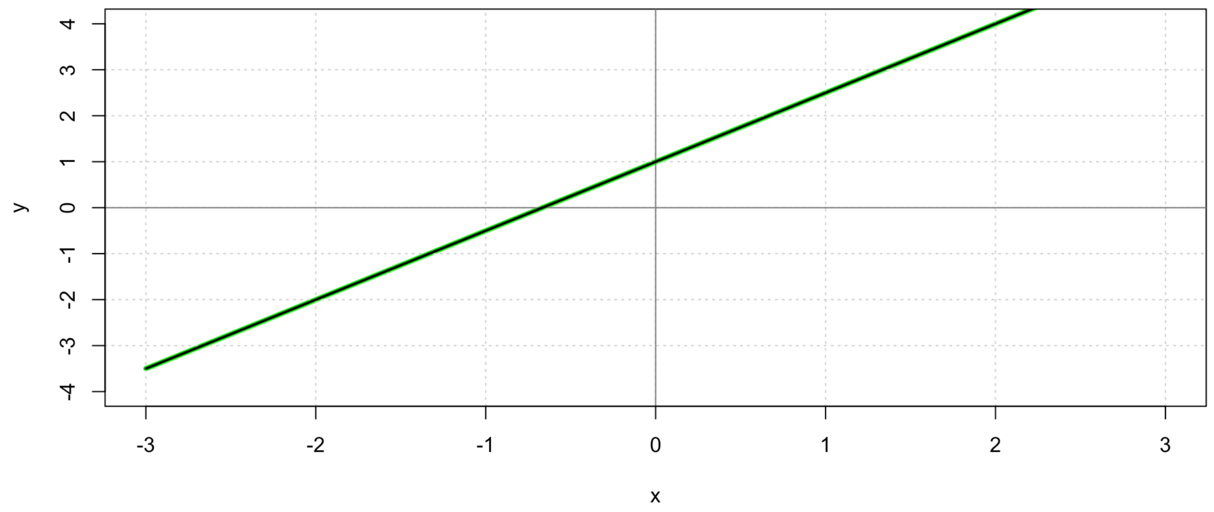
Equation of a line: practice

- Find the equation of the line that:

   a. passes through $(3,5)$ and $(2,-1)$.

   b. passes through $(2,-5)$ and has x-intercept equal to 4.

   c. has the following graph:



- Make a graph showing the line that passes through (3,-1) and has slope equal to 2.

Equation of a line: solutions

Find the equation of the line that passes through $(3, 5)$ and $(2, -1)$.

✏️ Let's see a solution.

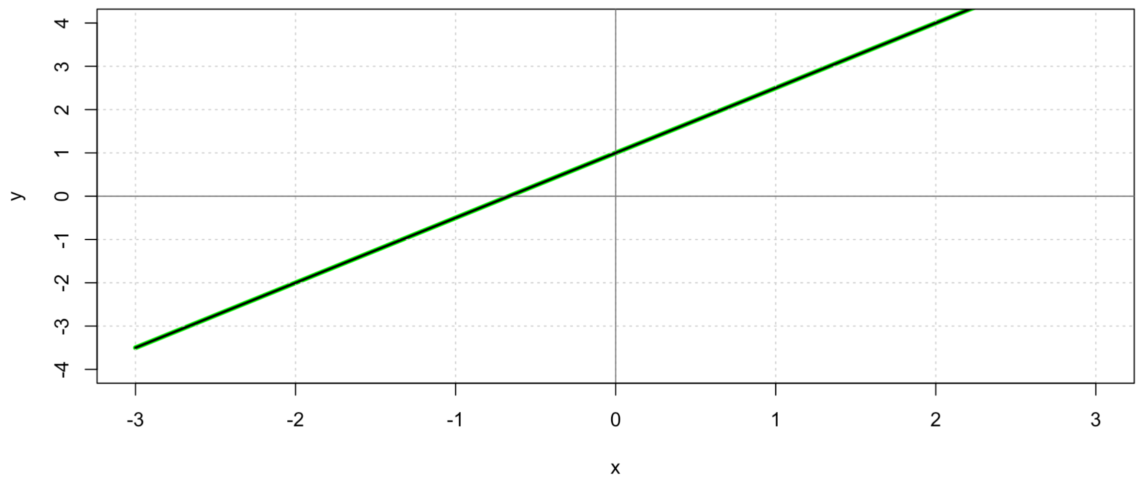
Equation of a line: solutions

Find the equation of the line that passes through $(2, -5)$ and has x-intercept equal to 4.

✏️ Let's see a solution.

Equation of a line: solutions

Find the equation of the line that has the following graph:



✏️ Let’s see a solution.

Equation of a line: solutions

Make a graph showing the line that passes through (3,-1) and has slope equal to 2.

✏️ Let's see a solution.

Interpreting the slope

Slope (average)

Sometimes, it can be useful to find the **average rate of change** of a function.

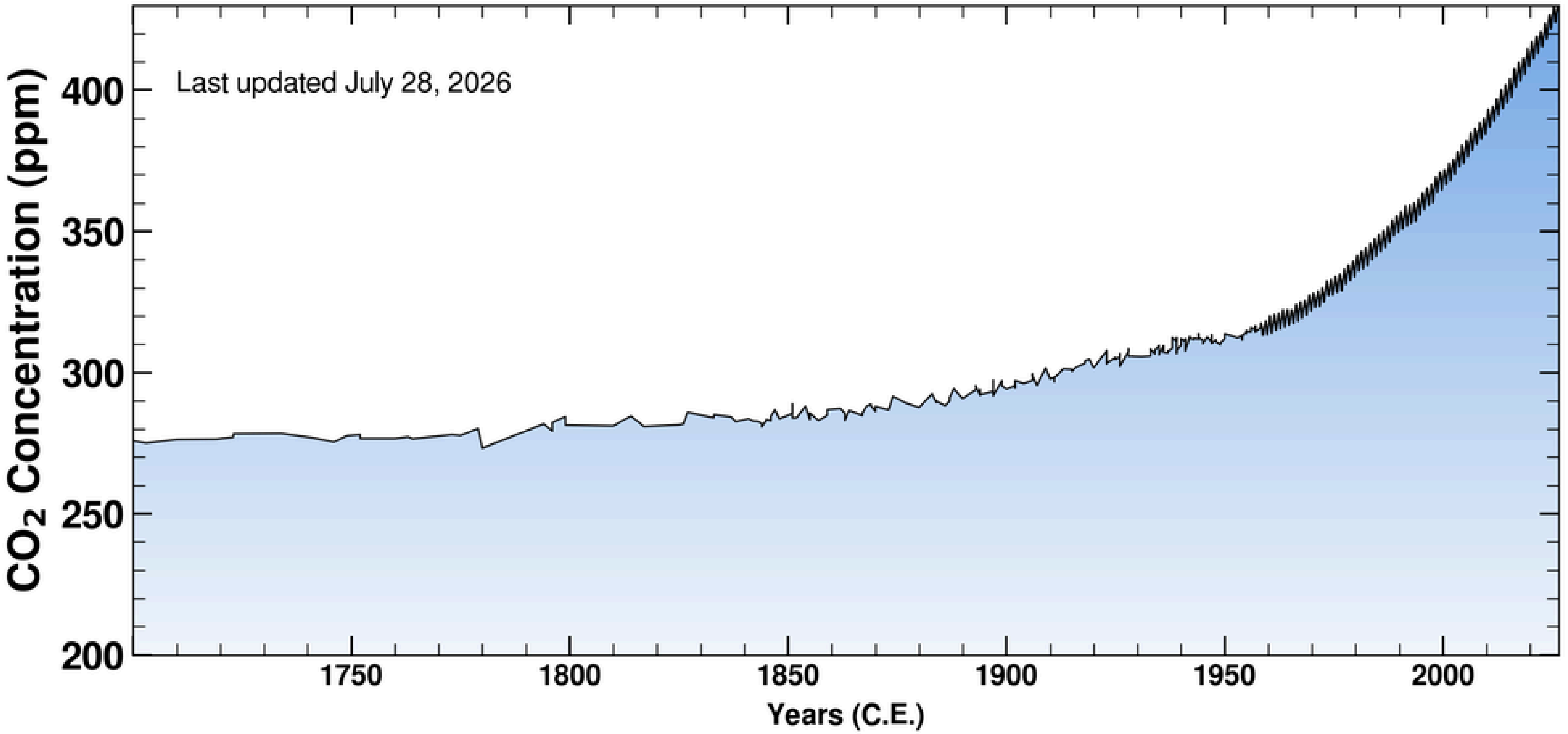
Between any two points (x_1, y_1) and (x_2, y_2) on the graph of a function, the slope is found by:

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Same idea as in a line!

Slope represents rate of change

Ice-core data before 1958. Mauna Loa data after 1958.



Source:The Keeling Curve

Get into the practice of saying the meaning out loud

- As if you're explaining it to someone unfamiliar with the data
- Including units
- Without overstating certainty

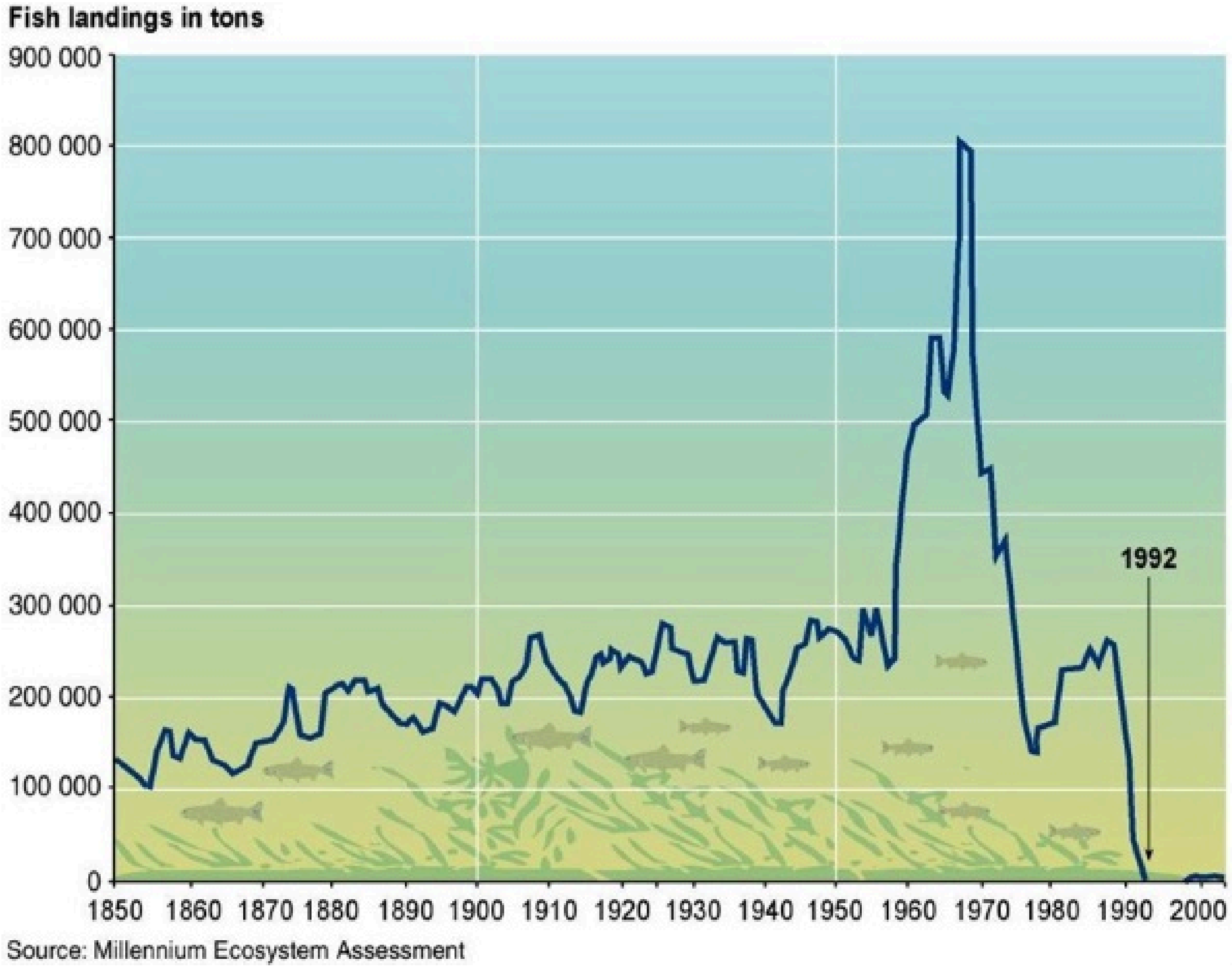
For example:

> "Between 1972 and 2020 the price of hobbit homes increased by an average of $2,450 per year"

differs from

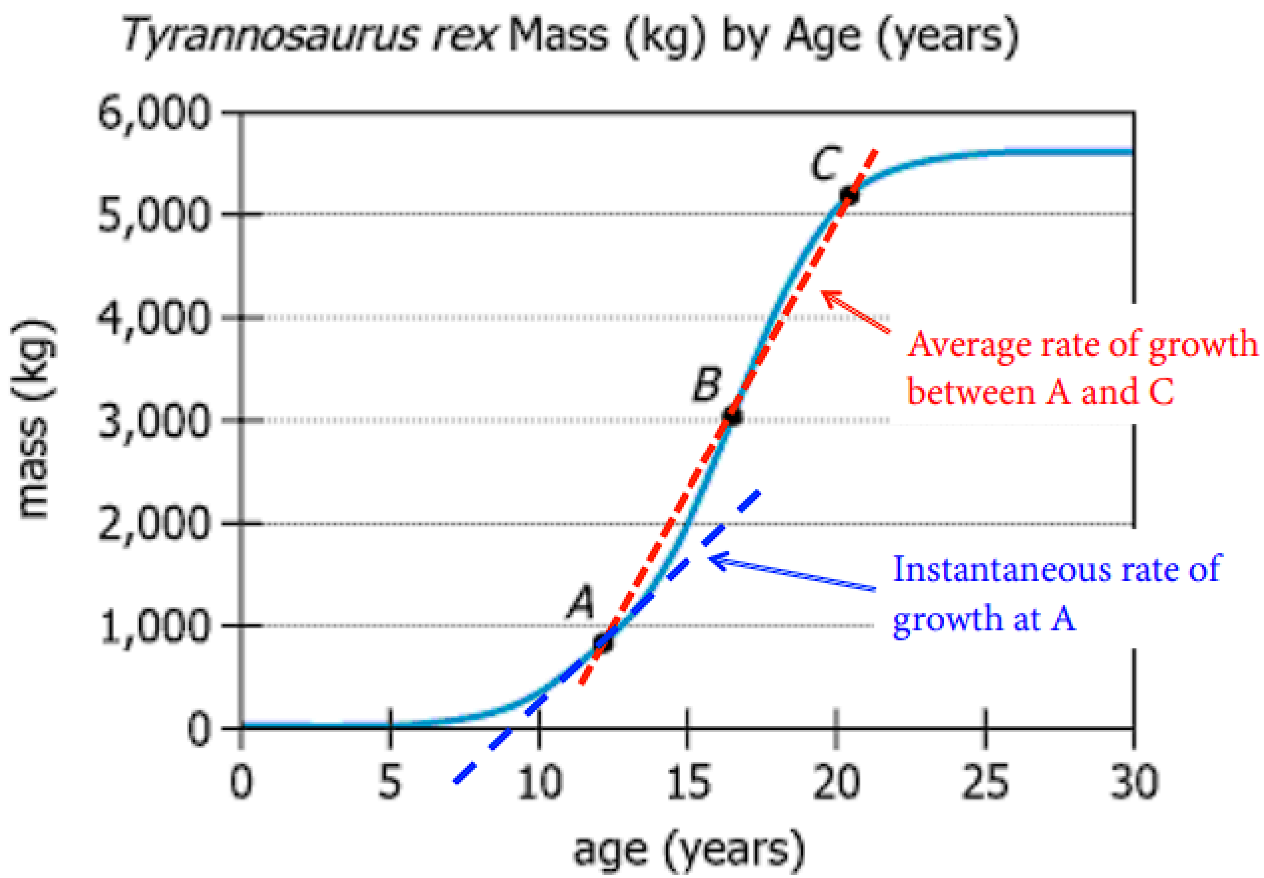
> "Between 1972 and 2020 the price of hobbit homes increased by $2,450 per year."

The *average* slope of a continuous function rarely tells the whole story



Slope can be *average* or *instantaneous*

- Rise over run can always be used to find average rate of change between two parts of a graph
- Instantaneous rate of change leads us to ✨calculus✨.



End of day 1!

What we covered today

- 🔢 Algebra review
- 🗺️ Solving equations
- ✖️ Exponents
- 📐 Polynomials
- 🌡️ Units
- 🧩 Definition of functions
- 📊 Graphing
- 💬 Reading graphs
- 📈 Linear functions
- ↗️ Slope as a rate of change